Injera Making Machine - System Architecture

Version: 1.0
Status: Draft
Target: Revision A (Standalone Controller)

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1. Purpose

This document defines the overall system architecture of the Injera Making Machine.

It describes:

- Product architecture
- Controller architecture
- Hardware architecture
- Software architecture
- Deployment architecture
- Scalability strategy

This document is the highest-level technical document for the project.

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2. Design Philosophy

The project shall be designed as a configurable automation platform rather than a single-purpose machine.

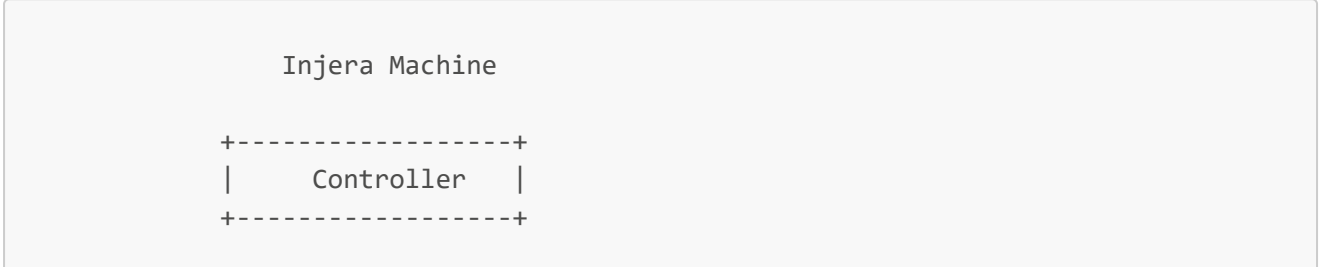
The same hardware and software platform shall support:

- Standalone machine
- Sequential multi-plate machine
- Parallel multi-plate machine

through configuration rather than redesign.

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3. Product Architecture



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     +-------+-------+-------+-------+
     |       |       |       |       |
  Motion  Heating  Dispense Inputs Outputs
```

The controller coordinates all machine subsystems.

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4. Deployment Modes

Standalone

One controller operates one baking station.

```
Controller

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One Plate
```

Sequential

One controller operates multiple stations sequentially.

A shared dispensing mechanism is transferred between plates.

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Plate 1

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Plate 2

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Plate 3
```

Transfer motion is enabled.

Parallel

Each station operates independently.

```
Station 1

Station 2

Station 3
```

Transfer motion is disabled.

Each station has its own dispensing mechanism.

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5. Controller Architecture

The controller consists of the following modules.

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Power

Digital Inputs

Digital Outputs

Motion Control

Temperature

MCU Interface

Safety
```

Each module has an independent schematic sheet.

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6. Motion Architecture

The controller supports two configurable motion channels.

Axis A

Purpose:

Dispensing rotation (Theta).

Required in every deployment.

Axis B

Purpose:

Transfer motion.

Examples:

- Shared dispenser movement
- Plate indexing
- Future expansion

Optional.

The firmware determines whether Axis B is used.

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7. MCU Interface

The controller shall not depend on a specific development board.

The MCU Interface is the contract between the controller hardware and the processor.

Revision A:

STM32 NUCLEO-F446RE

Revision B:

STM32F446RE directly mounted on PCB

Future:

Other STM32 families may implement the same interface.

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8. Hardware Layers

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Application

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Machine Functions

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MCU Interface

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STM32 HAL
```

```
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Hardware
```

9. Software Layers

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Application

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Control System

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Motion

Thermal

Pump

Safety

↓

MCU Interface

↓

STM32 HAL
```

10. Safety

Safety shall always be independent from normal machine operation.

Safety functions include:

- Emergency stop
- Cover interlock
- Over-temperature
- Watchdog

Hardware safety shall always have priority over software.

11. Scalability

The architecture shall support:

Standalone

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Sequential

↓

Parallel

↓

Industrial Network

without redesign of the controller PCB.

12. Revision Strategy

Revision A

- NUCLEO-based
- Single controller
- Standalone validation
- Production-quality controller PCB

Revision B

- STM32 on PCB
- Optimized BOM
- Industrial enclosure

Revision C

- Multi-controller networking
- Industrial deployment
- Factory integration

13. Guiding Principles

1. One hardware platform
2. One firmware platform
3. Configurable deployment

4. Modular architecture
5. Design for debugging
6. Design for manufacturing
7. Design for scalability